

Non-Conversion Web Campaign Attribution

This doc outlines how we track non-conversion events on the web as conversions using Munchkin activities and Workato-powered Marketo field updates.

Overview

What it does: Automatically tracks content downloads (PDFs, whitepapers, case studies) and other micro-conversions by logging Munchkin activities and updating Marketo lead fields via Workato.

Why it matters: By capturing these interactions, we can:

- Track engagement from both anonymous and known visitors
- Attach conversions to Salesforce Campaigns
- Track them in our core attribution system
- Measure content effectiveness
- Attribute pipeline and revenue to specific marketing touchpoints

User experience: PDFs open instantly in new tabs. No forms to fill out. No delays. Seamless tracking happens invisibly in the background. Can also be attached to any other Google Tag Manager event.

Business value: Every content download becomes a trackable, attributable marketing touchpoint that flows into Salesforce, enabling accurate ROI measurement and multi-touch attribution.

What's New in v3.0

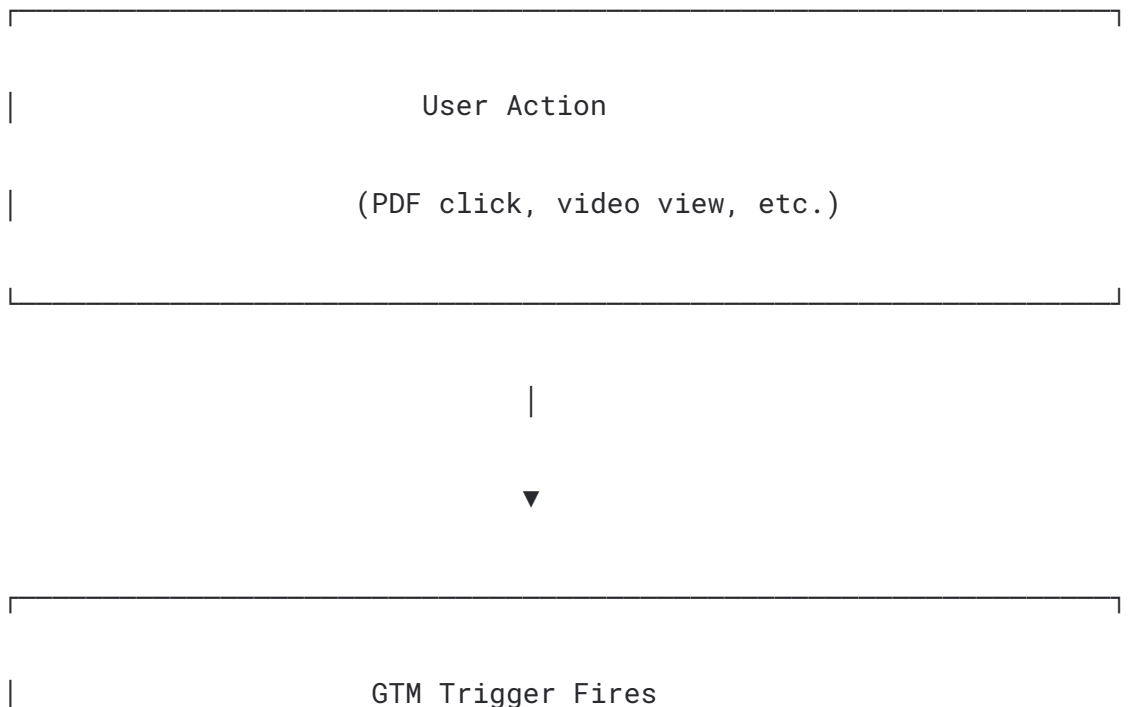
Previous Approach (v2.x) - Hidden Marketo Form

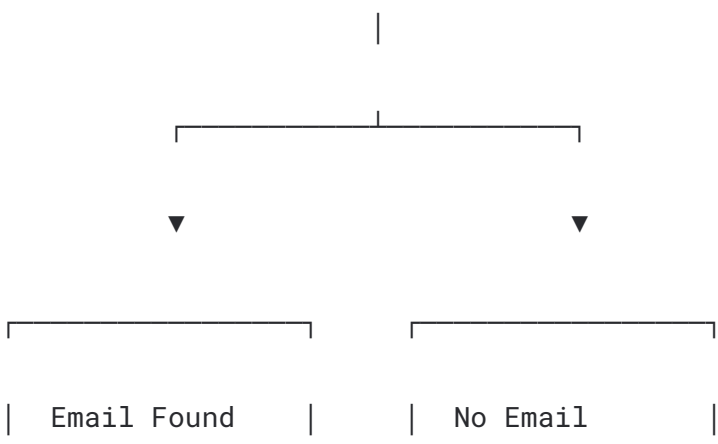
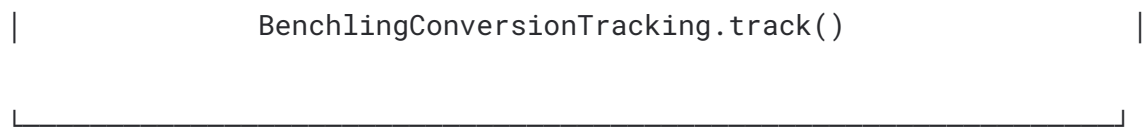
- Loaded a hidden Marketo form (ID: 1712) on every page
- Submitted form in background with UTM data
- **Problem:** CORS errors when other visible Marketo forms existed on the same page
- **Problem:** Marketo's MktoForms2 library uses a single shared iframe that conflicts when multiple forms are present

New Approach (v3.0) - Munchkin + Workato Hybrid

- **Anonymous visitors:** Munchkin `visitWebPage` activity logs conversion with UTM data encoded in URL
 - **Known visitors:** Munchkin activity + Workato webhook updates Marketo lead fields directly via REST API
 - **Benefit:** No CORS issues, works on all pages regardless of other forms
 - **Benefit:** Anonymous conversions are preserved and associate when visitor becomes known
-

Architecture





(Known User)	(Anonymous)
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▼	▼
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Workato Webhook	Done
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POST with UTMs	(Munchkin only)
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▼

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Marketo REST API

Update Lead

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▼

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Salesforce Sync

What This Script Does

The Problem It Solves

Traditional approach:

User clicks PDF → Lands on
form page → Fills out form →
Gets PDF

Issues:

- High friction (form fill required)
- Abandoned downloads (users leave)
- Can't track anonymous/known visitors who don't fill forms
- Poor user experience

Our solution:

User clicks PDF → PDF opens immediately in new tab

→ Background: Munchkin logs activity

→ If known: Workato updates Marketo fields

→ Conversion tracked with full attribution data

→ Flows to
Salesforce campaign

Benefits:

- Zero friction (no form to fill)
 - Track known visitors automatically (via cookie email)
 - Track anonymous visitors (activity associates when they become known)
 - Instant access to content
 - Complete attribution chain maintained
 - Better user experience = more engagement
 - No CORS issues on pages with other Marketo forms
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Core Functionality

Dual-Track Recording

Visitor Type	Munchkin Activity	Workato → Marketo Fields
Anonymous	✓	—
Known	✓	✓

Munchkin Activity (All Visitors)

- Logs `visitWebPage` activity with conversion type in URL path
- UTM parameters encoded in query string
- Activity associates with lead when anonymous visitor becomes known
- Example: `/conversion/PDF%20Download?s=google&m=cpc&a=brand`

Workato Field Update (Known Visitors Only)

- Sends webhook with full attribution payload
- Workato searches for lead by email
- Updates UTM fields, conversion type, conversion URL
- Direct REST API call—no CORS issues

Attribution Data Capture

- UTM parameters (source, medium, campaign, term, content)
 - Google Ads data (creative, match type, device, location)
 - Referral URLs and landing pages
 - Conversion type and timestamp
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How It Works

Technical Flow

Step 1: Page Load

1. Script loads via GTM on all pages
2. BenchlingConversionTracking API becomes available
3. No forms to load—ready immediately

Step 2: User Action (e.g., Clicks PDF)

1. GTM trigger fires
2. Calls `BenchlingConversionTracking.track({ conversionType: 'PDF Download' })`

Step 3: Munchkin Activity

1. Script calls `Munchkin.munchkinFunction('visitWebPage', {...})`
2. Logs activity with conversion path and UTM params
3. Activity tied to Munchkin cookie (anonymous or known)

Step 4: Email Lookup

1. BenchlingDTO instantiated with callback
2. Cross-domain iframe retrieves email from Marketo cookie
3. If email found → proceed to Workato
4. If not found → done (Munchkin activity already logged)

Step 5: Workato Webhook (Known Only)

1. POST to Workato webhook URL with payload:

- email
 - conversion_type
 - conversion_url
 - All UTM parameters
 - Google Ads data
2. Workato receives webhook

Step 6: Marketo Update

1. Workato searches Marketo for lead by email
2. Updates lead fields with attribution data
3. Lead record reflects latest conversion

Step 7: Salesforce Sync

1. Marketo syncs updated fields to Salesforce
2. Available for reporting and attribution

Total time: 1-2 seconds, all happening invisibly.

Setup & Installation

Prerequisites

Required:

- Munchkin tracking implemented on site
- BenchlingDTO (cross-domain tracking) implemented
- Tracking cookies (`_bnchs`, `_bnchad`) in place
- Google Tag Manager
- Workato account with Marketo connector

Installation Steps

Step 1: Workato Recipe

Create webhook recipe:

1. **Trigger:** Webhooks by Workato → New request
 - Note the webhook URL generated
2. **Step 1:** Marketo → Search leads
 - Filter type: email
 - Email: [Trigger → email]
3. **Step 2:** Conditional → If lead exists
 - Condition: [Step 1 → Lead ID] is present
4. **Step 3 (Yes branch):** Marketo → Update lead
 - Lead ID: [Step 1 → id]
 - Field mappings:

Marketo Field	Workato Formula
utm_source__c	[Trigger → utm_source]
utm_medium__c	[Trigger → utm_medium]
utm_campaign__c	[Trigger → utm_campaign]
utm_content__c	[Trigger → utm_content]
utm_term__c	[Trigger → utm_term]
Last_Referral_Page_URL__c	[Trigger → last_referral_url].present? ? [Trigger →

	<pre>last_referral_url].split("?").first.split("#").first.first(255) : nil</pre>
Last_Landing_Page_URL__c	<pre>[Trigger → last_landing_url].present? ? [Trigger → last_landing_url].split("?").first.split("#").first.first(255) : nil</pre>
Google_Ads_Creative__c	<pre>[Trigger → gads_creative]</pre>
Google_Ads_Match_Type__c	<pre>[Trigger → gads_match_type]</pre>
Google_Ads_Device__c	<pre>[Trigger → gads_device]</pre>
Google_Ads_Location__c	<pre>[Trigger → gads_location]</pre>
Last_Campaign_Notes__c	<pre>[Trigger → conversion_type]</pre>
Last_Conversion_Page_URL__c	<pre>[Trigger → conversion_url].present? ? [Trigger → conversion_url].split("?").first.split("#").first.first(255) : nil</pre>

5.

Step 4 (No branch): Logger or Slack notification

- Message: "Lead not found: [Trigger → email]"

Step 2: Update Script Configuration

In the GTM script, update the webhook URL:

```
var
CONFIG
=
{

workatoWebhookUrl
:
'https://www.workato.com/webhooks/rest/YOUR_ACTUAL_WEBHOOK_ID'
,

// ... rest of config

}
;
```

Step 3: Deploy GTM Tag

1. Create new tag: Custom HTML
2. Paste the conversion tracking script
3. Trigger: All Pages
4. Publish

Step 4: Create GTM Triggers

For PDF downloads:

1. **Trigger:** Click - Just Links
 - Fires on: Some Link Clicks
 - Condition: Click URL ends with .pdf
2. **Tag:** Custom HTML

```
<
script
>
```

```
if (typeof BenchlingConversionTracking !== 'undefined') {

  BenchlingConversionTracking.track({ conversionType: 'PDF Download' });

}
```

```
</
script
>
```

Usage Examples

Example 1: PDF Downloads

GTM Trigger:

- Type: Click - Just Links
- Condition: Click URL ends with .pdf

GTM Tag:

```
<
```

```
script
>
```

```
if (typeof BenchlingConversionTracking !== 'undefined') {

  BenchlingConversionTracking.track({ conversionType: 'PDF Download' });

}
```

```
</
script
>
```

What happens:

1. User clicks PDF link
2. Munchkin logs `/conversion/PDF%20Download?s=...`
3. If known user, Workato updates Marketo fields
4. PDF opens normally

Example 2: Video Views

GTM Trigger:

- Type: Custom Event
- Event name: `video_progress`
- Condition: Video Percent `>= 75`

GTM Tag:

```
<
script
>
```

```
if (typeof BenchlingConversionTracking !== 'undefined') {
```

```
BenchlingConversionTracking.track({ conversionType: 'Video View - 75%'
});

}

</script>
```

Example 3: Form Submissions

GTM Trigger:

- Type: Form Submission
- Condition: Form ID equals `contact-form`

GTM Tag:

```
<script>

if (typeof BenchlingConversionTracking !== 'undefined') {

    BenchlingConversionTracking.track({ conversionType: 'Contact Form
Submission' });

}

</script>
```

Example 4: Time on Page

GTM Trigger:

- Type: Timer
- Interval: 120000 (2 minutes)
- Limit: 1

GTM Tag:

```
<script>

if (typeof BenchlingConversionTracking !== 'undefined') {

  BenchlingConversionTracking.track({ conversionType: 'High Engagement - 2min' });

}

</script>
```

GTM Integration

Template for Any Conversion Type

```
<script>

(function() {
```

```

if (typeof BenchlingConversionTracking !== 'undefined') {

  BenchlingConversionTracking.track({

    conversionType: 'YOUR_CONVERSION_TYPE'

  });

}

})();

</script>

```

Best Practices

Name conversion types consistently:

- Use: "PDF Download - [Title]"
- Not: "pdf_download" or "PDF" or "download"

Use descriptive names:

- Good: "Webinar Registration - AI in Drug Discovery"
- Bad: "Form Submit"

Test in Preview mode first:

- Always test before publishing
- Check console for errors
- Verify in Marketo Activity Log
- Check Workato job history

Document your conversion types:

- Keep a list of all conversion types
- Share with ops and analytics teams

- Use consistent naming across campaigns
-

Troubleshooting

Common Issues

Issue: Tracking not firing

Symptoms:

- No Munchkin activity logged
- No Workato job triggered

Diagnosis:

```
// In console:
```

```
console
.
log
(
  typeof
  BenchlingConversionTracking
)
;
```

```
// Should output: "object"
```

```
BenchlingConversionTracking
.
enableDebug
```

```
(  
)  
;
```

```
BenchlingConversionTracking
```

```
.
```

```
track
```

```
(
```

```
{
```

```
conversionType
```

```
:
```

```
'Test'
```

```
}
```

```
)
```

```
;
```

```
// Watch console for log  
messages
```

Fix:

- Check if script loaded in GTM
- Check for JavaScript errors in console
- Verify GTM trigger is firing

Issue: Munchkin activity not appearing

Symptoms:

- Debug shows "Munchkin not available"

Diagnosis:

```
// In console:
```

```
console
```

```
.  
log  
(  
  typeof  
  Munchkin  
)  
;
```

```
// Should output: "object"
```

Fix:

- Verify Munchkin is loaded on the page
- Check Munchkin initialization

Issue: Workato not receiving webhooks (known users)

Symptoms:

- Debug shows "Anonymous visitor, Munchkin only"
- But user should be known

Diagnosis:

```
// In console:
```

```
BenchlingConversionTracking  
.  
enableDebug  
(  
)  
;
```

```
BenchlingConversionTracking
.
track
(
{
conversionType
:
'Test'
}
)
;

// Look for "BenchlingDTO
callback received" message
```

Fix:

- Verify BenchlingDTO is loaded
- Check if user has `_bnchs` cookie
- Verify cross-domain tracking iframe is accessible

Issue: Marketo fields not updating**Symptoms:**

- Workato job runs successfully
- But lead fields unchanged

Diagnosis:

- Check Workato job details for field mapping errors
- Verify field API names match Marketo

Fix:

- Confirm field API names (case-sensitive)

- Check field permissions in Marketo
- Verify lead exists with that email

Issue: Lead not found in Workato

Symptoms:

- Workato logs "Lead not found"
- But user filled out form previously

Possible causes:

- Lead was deleted or merged
- Email mismatch (different email used)
- Sync delay between systems

Enable Debug Mode

// In browser console:

```
BenchlingConversionTracking
.
enableDebug
(
)
;
```

Then trigger the action and watch console logs.

Health Check

Run this in console to verify setup:

```
console
.
log
(
```

```
'=== HEALTH CHECK ==='
```

```
)
```

```
;
```

```
console
```

```
.
```

```
log
```

```
(
```

```
'API loaded:'
```

```
,
```

```
typeof
```

```
BenchlingConversionTracking
```

```
!==
```

```
'undefined'
```

```
)
```

```
;
```

```
console
```

```
.
```

```
log
```

```
(
```

```
'Munchkin loaded:'
```

```
,
```

```
typeof
```

```
Munchkin
```

```
!==
```

```
'undefined'
```

```
)
```

```
;
```

```
console
.
log
(
  'BenchlingDT0 loaded:'
,
  typeof
  BenchlingDT0
  !==
  'undefined'
)
;
```

```
console
.
log
(
  'Config:'
,
  BenchlingConversionTracking
  &&
  BenchlingConversionTracking
  .
  getConfig
  (
  )
)
;
```

```
console
.
log
(
  'Session cookie:'
,
document
.
cookie
.
includes
(
  '_bnchs'
)
?
'Present'
:
'Missing'
)
;
```

```
console
.
log
(
  'Ads cookie:'
,
document
```

```
.
cookie

.
includes
(
  '_bnchad'
)
?
'Present'
:
'Missing'
)
;
```

Expected output:

=== HEALTH CHECK ===

API loaded: true

Munchkin loaded: true

BenchlingDTO loaded: true

Config: {workatoWebhookUrl: "https://...", debug: false, ...}

Session cookie: Present

Ads cookie: Present

Workato Webhook Payload

The script sends this JSON payload to Workato:

```
{  
  
  "email"  
  :  
  "user@example.com"  
  ,  
  
  "conversion_type"  
  :  
  "PDF Download"  
  ,  
  
  "conversion_url"  
  :  
  "https://www.benchling.com/re  
sources/whitepaper.pdf"  
  ,  
  
  "conversion_timestamp"  
  :  
  "2026-01-06T12:34:56.789Z"  
  ,  
  
  "utm_source"  
  :  
  "google"  
  ,  

```

"utm_medium"

:

"cpc"

,

"utm_campaign"

:

"brand"

,

"utm_content"

:

" "

,

"utm_term"

:

" "

,

"last_referral_url"

:

"google.com"

,

```
    "last_landing_url"  
    :  
    "www.benchling.com/demo"  
    ,  
  
    "gads_creative"  
    :  
    ""  
    ,  
  
    "gads_match_type"  
    :  
    ""  
    ,  
  
    "gads_device"  
    :  
    ""  
    ,  
  
    "gads_location"  
    :  
    ""  
  
}
```

FAQ

General Questions

Q: Does this work for anonymous users?

A: Partially. Anonymous users get a Munchkin `visitWebPage` activity logged. When they later become known (fill out a form), that activity associates with their lead record. However, UTM field updates only happen for known users via Workato.

Q: Will users notice any difference?

A: No. All tracking happens invisibly in the background. No forms, no delays.

Q: What if the user blocks cookies?

A: Tracking will be limited. Munchkin requires cookies. This affects ~5-10% of users with strict privacy settings.

Q: Can I track non-PDF content?

A: Yes. Create GTM triggers for any event and call `BenchlingConversionTracking.track()` with the appropriate `conversionType`.

Q: What happened to the hidden Marketo form?

A: Removed in v3.0 due to CORS errors. When multiple Marketo forms exist on a page, they share a single cross-domain iframe that causes submission conflicts. The new Workato approach uses REST API calls which don't have this limitation.

Technical Questions

Q: What if Workato is slow or unavailable?

A: The script fires Munchkin activity immediately (doesn't wait). Workato webhook is fire-and-forget. User experience is never blocked.

Q: Can I track the same user multiple times?

A: Yes. Each conversion triggers a new Munchkin activity and Workato call. Marketo fields get overwritten with the latest attribution data.

Q: How do I track different conversion types?

A: Create different GTM triggers and tags, each calling `track()` with a different `conversionType` value.

Q: What happens if the script fails?

A: User action proceeds normally. Tracking fails silently. No errors shown to user.

Q: Does this affect page load speed?

A: Minimal impact. The script is lightweight (~5KB) and loads asynchronously. No external libraries to load (Munchkin is already on the page).

Q: Why both Munchkin AND Workato?

A: Munchkin captures the activity for anonymous visitors (associates when they become known). Workato updates lead fields immediately for known visitors so attribution data is current.

Migration from v2.x

If upgrading from the hidden form approach:

1. **Remove old script** from GTM
2. **Deploy new script** with Workato webhook URL configured
3. **Create Workato recipe** as documented above
4. **Test** with debug mode enabled
5. **Verify** in Marketo Activity Log and Workato job history

The hidden Marketo form (ID: 1712) is no longer used and can be archived.

Version History

Version	Date	Changes
3.0.0	January 2026	Replaced hidden Marketo form with Munchkin + Workato hybrid approach. Fixes CORS issues on pages with visible forms.
2.1.0	October 2025	Hidden Marketo form approach with click interception

Version: 3.0.0

Last Updated: January 2026

Maintained By: Marketing Operations